Termal

User Manual

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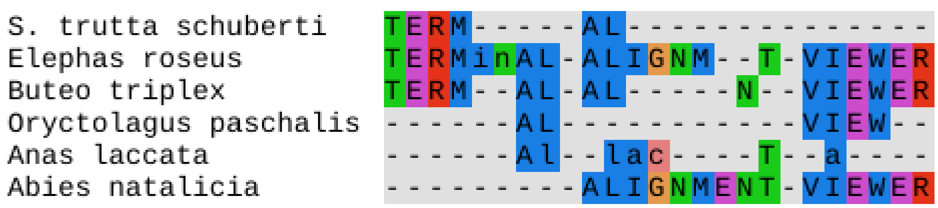

Contents

Overview

Termal is a terminal-based viewer for multiple sequence alignments (MSAs). It is designed for fast, keyboard-driven navigation of large alignments, particularly in remote or SSH-based environments where graphical tools are impractical.

Termal is a **read-only** viewer. It does not modify alignments. This may change in future versions.

Termal is presented in (Junier 2025).

Basic usage

Starting Termal

Simply pass your alignment file as an argument to `termal`, e.g.:

```
termal my-alignment.msa
```

Alignment File Formats

Termal supports Fasta (default) and Stockholm (pass `-f stockholm` (or just `-f s`)) formats. Any alignment lines shorter than the longest one will be padded with gap characters at the end.

Help

For general help, pass option `-h`; for a list of key bindings, pass `-b` or press `?` after launching `termal`.

Info mode

With option `-i`/`--info`, `termal` will output basic metrics about the alignment (such as number of sequences) and quit.

Options

The main options are shown in the table below (and are discussed in the text in the corresponding sections). Other options exist, but they are either experimental or used for debugging or testing. For a full list, do `termal -h`.

short	long	function
`-b`	`--show-bindings`	Show key bindings and exit successfully
`-i`	`--info`	Show alignment metrics and quit
	`--citation`	Show citation information and exit successfully
`-f`	`--format <FORMAT>`	Sequence file format [`fasta`/`stockholm`] (or just `f`/`s`); default: `fasta`
`-o`	`--user-order <USER_ORDER>`	User-supplied order (filename)
`-c`	`--color-map <COLOR_MAP>`	Gecos color map (filename)
`-n`	`--dry-run`	Dry run - show parameters and quit
`-h`	`--help`	Print help
`-V`	`--version`	Show version

Screen layout

The interface is divided into four areas, starting from the top and left to right::

- **Headers pane** or simply **left pane**: shows sequence numbers and headers, as well as a barplot of the current metric
- **Alignment pane** or **main pane**: shows the aligned sequences, some properties of the alignments, as well as some current UI settings.
- **Corner pane**: shows the current metric and ordering (see below).
- **Reference pane** or **bottom pane**: shows horizontal position, the reference sequence (usually the consensus, but see setting the reference), and a conservation barplot

In addition, the last line contains a message area ("modeline"), which displays:

- pending command arguments (counts, searches), if any;
- search match information (when applicable).

Interaction

Termal is entirely keyboard-driven. All commands are issued by typing a single character (usually a letter).

Prefix arguments

Many commands (mostly motion) accept a *prefix argument*, which is an integer number typed *before* the command character. Typing any digit (except 0) starts a prefix argument. To cancel, press `Esc`. The meaning of the argument depends on the command. For example, scrolling commands such as `j` ("scroll one line down") interpret a prefix argument as a repetition count: typing just `j` will move one sequence down, but typing `12j` will move twelve sequences down.

If no prefix argument is given, commands default to **1**.

String Arguments

String arguments are entered after typing the command character, and are entered by typing `Return/Enter`. Currently only the header search command (`"`), sequence search command (`/`) and reference selection command (`R`) take a string argument.

Navigation fundamentals

Navigation in Termal is inspired by Vim but simplified and adapted to multiple sequence alignments. The effect of motion commands depends on the zoom mode, as follows:

- In zoomed-in mode, they change the portion of the alignment that is being displayed, much as a pager pages through a text file.
- In the zoomed-out modes, they move the zoom box.

Motion commands all take a prefix argument.

Scrolling

Scrolling commands move the visible part of the alignment by one or more steps. The prefix argument is interpreted as a repeat count.

Single-step scrolling

The following keys move the viewport by one step.

command	motion
`[count]h`	scroll *count* columns to the left
`[count]l`	scroll *count* columns to the right
`[count]j`	scroll *count* sequences down
`[count]k`	scroll *count* sequences up

Scrolling by screenfuls

To move by a screenful, use uppercase motion commands:

command	motion
`[count]H`	scroll *count* screenfuls to the left
`[count]L`	scroll *count* screenfuls to the right
`[count]J`	scroll *count* screenfuls down
`[count]K`	scroll *count* screenfuls up

Aliases

The arrow keys are aliases for the motion keys.

arrow	alias for
←	h
→	l
↓	j
↑	k
Shift-←	H
Shift-→	L
Shift-↓	J
Shift-↑	K

The Space key is an alias for J (move one screen down). This mimics the behaviour of less and other Unix pagers.

Examples

- h: move 1 column to the left
- 10l: move 10 columns to the right
- 5K: move 5 screens up
- 2<Spc>: move two screens down

Jumps

Jump commands move directly to a position, specified by the prefix argument. The viewport moves so that the target sequence (respectively column) appears at the top (respectively left) of the alignment pane.

NOTE: in vertical jumps, the sequence's number is counted from the *first screen line*. This will be the same as the sequence's number in the alignment file, unless the sequences have been reordered.

Absolute jumps

In absolute jumps, the prefix argument denotes a specific sequence or column:

command	motion
[count]-	jump to sequence *count* (in the current order)
[count]=	jump to sequence *count* (in original file order)
[count]|	jump to column *count*

Relative jumps

In relative jumps, the prefix argument denotes a percentage of the alignment's height or width, rounded to the nearest sequence/column.

command	motion
[count]%	jump to *count* % of the alignment's height
[count]#	jump to *column* % of the alignment's width

Jumps to search matches

Termal supports regular expression searches in both headers and sequences (see Searching for how to start a search). If any matches are found, Termal will automatically jump to the first match. To jump to the next or previous match, use n or p:

command	motion
`[count]n`	jump *count* matches forward
`[count]p`	jump *count* matches backward
`<Return>`	jump to the current match (which may be offscreen)

Next and previous match jumps wrap around, i.e. pressing `n` while on the last match will move back to the first one. If no matches were found, match jump commands have no effect.

Examples

- `123|`: Jump to column 123 (if it exists).
- `200-`: Jump to the 200th sequence from the top of the display, which may differ from the original file order if sequences have been reordered.
- `50%`: Jump to the vertical midpoint of the alignment.
- `25#`: Jump to one quarter of the alignment width.
- `n`: jump to the next match
- `3p`: jump three matches backward

Zooming

In Termal, *zooming* refers to changing how the alignment is displayed, rather than scaling characters graphically (although this can be done by changing the terminal emulator's font size). Zooming is only relevant if the whole alignment does not fit on screen. It is possible for the alignment to fit on the screen only in one of the two dimensions: in this case, the zooming only applies to the other dimension.

In *zoomed-in* mode, Termal displays as many *adjacent* sequences and columns as will fit on the screen. This shows a detailed view of a portion of the alignment, but obscures its large-scale features. Motion commands change which portion of the alignment is displayed, causing new sequences/columns to appear into view and as many sequences/columns to disappear. Termal starts in zoomed-in mode.

In *zoomed-out* mode, Termal shows the first and last sequence/column as well as a uniform sampling of sequences/columns between them. Enough sequences/columns are sampled to fill the screen; the other sequences/columns are not shown. This shows a "big picture" view of the alignment, at the expense of granularity. Motion commands affect the part of the alignment that will be shown when moving back to zoomed-in mode. This area is shown on screen as a rectangle known as the *zoom box*.

A variant of the *zoomed-out* mode works as above but preserves the alignment's aspect ratio. It is called *zoomed-out-AR*.

To cycle forward through the zoom modes, press `z`; to cycle backward, press `Z`.

Searching

The Search Prefix Command

All searches can be initiated via the `f` prefix command:

command	action
`fh`	header search (see Searching sequence headers)
`fs`	sequence search, ignoring gaps (see Searching sequences)
`fa`	alignment search: matches the raw alignment row, including gap characters

The standalone " and / commands are shortcuts for fh and fs respectively. fa (alignment search) is only available through the f prefix.

Searching sequence headers

Termal supports searching within sequence headers using regular expressions.

To start a search:

1. Press " (double quote)
2. Enter a regular expression
3. Press <Enter> to confirm

To cancel a search, press <Esc>.

During a search, the modeline displays:

- the index of the current match,
- the total number of matches.

Example:

```
"^Eco<Enter>
```

This jumps to the first sequence whose header starts with Eco (if any). The search order is according to the current ordering.

Searching sequences

Termal supports searching within sequences using regular expressions.

To start a search:

1. Press /
2. Enter a regular expression
3. Press <Enter> to confirm

To cancel a search, press <Esc>.

During a search, the modeline displays:

- the index of the current match,
- the total number of matches.

NOTE By default, sequence search ignores gap characters: the regular expression is matched against the ungapped sequence, and match positions are mapped back to the alignment. To search the raw alignment row including gaps (e.g. to match A-?T literally), use the alignment search command fa instead (see Searching).

Example:

```
/GAATTC<Enter>
```

This jumps to the first instance of GAATTC. The search order is according to the current ordering, then left to right.

Setting the Reference

Some features of the alignment are computed with respect to a *reference sequence*. This is the case, for example, of the similarity metric, which measures how similar each sequence is to the reference.

By default, the reference is the consensus sequence (which is automatically computed). However, the user may select any sequence in the alignment to serve as reference. This is done with the `R` command, which takes a sequence number (with respect to the original file) as an argument. If the sequences are currently ordered according to the similarity metric, changing the reference triggers a recomputation of the ordering.

Passing an empty argument reverts to the consensus.

Example:

```
R12<Enter>
```

This selects sequence #12 as reference. This is reflected in the corner pane, which now reads 'Ref: #12', and in the bottom pane, which displays sequence #12 instead of the consensus.

Display Controls

Resizing the Left Pane

The left pane can be widened (perhaps to show more of the sequence headers) with `>` and shrunk with `<`. This also resizes the corner pane. Both accept a prefix argument, which is by how many characters the pane is to be resized:

command	motion
`[count]>`	widen the left pane by *count* characters
`[count]<`	shrink the left pane by *count* characters

Ordering the Sequences

By default, the sequences appear in the alignment pane in the same order as they appear in the alignment file. However, the sequences may be ordered according to the current metric, either ascending or descending. Note that the "first" sequence (on the screen) is the *top* sequence. As a result, when sorting in ascending order, metric values increase from top to bottom.

The user may also supply a custom ordering, by passing option `-o` and supplying the name of an ordering file (see below) as an argument to the option, e.g.:

```
termal -o my-order alignment.msa
```

In this case, `termal` initially shows the alignment in custom ordering.

Ordering File

An ordering file is simply the sequence headers in the desired order. For example, passing the following ordering file would cause `AHMKMHDK_00298` to appear first, then `CEFNMEKK_03699`, etc., regardless of the order they appear in in the alignment file.

Note that the headers in the ordering file are expected to match those in the alignment file.

```
AHMKMHDK_00298
CEFNMEKK_03699
IAGGDKJC_03995
```

```
FHLODNDD_02091
HJACHOPP_04370
```

Changing the Ordering

To cycle forward through orderings, press (`o`). To cycle backward, press `O`.

The current ordering is displayed is the corner pane, as a symbol just below the bar chart in the left pane:

symbol	meaning
-	file order
↑	current metric, ascending
↓	current metric, descending
u	user-supplied order (`-o`)

Metrics

The left pane displays a bar chart of the current *metric*. This is a numeric property of the sequences. Currently, there are two possible metrics:

a. Sequence length (not counting gaps)
b. Similarity to the reference

To cycle forward through the metrics, press `t` (me**t**ric); press `T` to cycle backward. The current metric is displayed in the corner pane.

The sequences can be ordered according to the current metric.

Column Metrics

The bottom pane displays a barplot of the current *column metric*, a numeric property of each alignment column. Two metrics are available:

a. **Entropy**: a measure of conservation. High bar = well-conserved column; low bar = high variability.
b. **Coverage**: fraction of non-gap residues at that column.

To cycle through column metrics, press `c` (forward) or `C` (backward). The current metric name is shown in the bottom pane.

Residue Colormaps

Termal supports four built-in residue color maps:

source	residue class	reference
ClustalX	amino acids	(Larkin et al. 2007)
Lesk	amino acids	(Lesk 2019)
JalView	nucleotides	(Waterhouse et al. 2009)
monochrome	both	themes

It is also possible to supply a custom color map:

```
termal -c colormap.json my-alignment.msa
```

where the colormap is a JSON file in the Gecos format (Kunzmann et al. (2020)). This is a straightforward format that looks like this:

```json
{
    "name": "my-colormap",
    "alphabet": [
        "A",
        "C",
        "G",
        "T",
    ],
    "colors": {
        "A": "#71564e",
        "C": "#2a3d00",
        "G": "#004f7c",
        "T": "#b03f42",
    }
}
```

To cycle through colormaps, type `m` (forward) or `M` (backward). The initial colormap is ClustalX for amino acids or JalView for nucleotides, unless a custom colormap is supplied, in which case it becomes the initial colormap. The current colormap is displayed in the top border of the Termal screen.

Themes

Termal supports three themes: dark, light, and monochrome. To cycle through the themes, type `s` (forward) or `S` (backward). The current theme is displayed in the top border of the Termal screen.

NOTE Termal has been tested predominantly in a dark-themed terminal.

Diff Mode

In diff mode, each residue in the alignment is displayed relative to the reference:

symbol	meaning
letter	residue differs from the reference at this position
.	residue is identical to the reference
-	actual gap in the sequence

This makes it easy to spot variation at a glance, especially when combined with setting a specific sequence as reference.

To enter diff mode, press `dd`; to return to normal mode, press `D` or `dn`. The reference sequence row is always shown in full, regardless of diff mode.

Inverse Video

By default, Termal displays the sequence residues in inverse video. To toggle to direct video (and back), press `i`. The current video mode is displayed in the top border of the Termal screen.

Modeline and feedback

The modeline (in the bottom border of the Termal screen) provides feedback about: - pending numeric arguments, - active search state, - current search match index.

Limitations and scope

Termal currently does **not** support:

- editing alignments
- graphical export
- alignment formats other than Multi-fasta and Stockholm

Future Work

Features planned for the next release:

- jumps to conserved regions
- a color map for vision-impaired users

Features that will be added later

- exporting (part of) the alignment as SVG
- showing fully conserved residues (à la EMBOSS's `showalign -show`)

Features under consideration

- Showing a phylogeny in the left pane
- Simple edits (removing empty columns)
- Saving parts of the alignments

Design philosophy

Termal prioritizes:

- predictability over feature breadth,
- keyboard-driven navigation over menus,
- robustness over visual effects.

It is intended for users comfortable with terminal-based tools who value speed and clarity over graphical interaction.

Many commands, as well as the prefix argument syntax, were deliberately copied from Vim.

References

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